Nan Xiao

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Overview

AI Engineer with a background in Applied Mathematics and Statistics, experienced in building production-grade LLM systems, multi-agent evaluation platforms, and scalable AI infrastructure. Proficient in bridging machine learning experimentation with distributed AI systems, including multi-agent orchestration, evaluation pipelines, async inference, and observability infrastructure.

Technical Skills

Languages: Python, SQL, Java, TypeScript/JavaScript, HTML/CSS
AI / ML: PyTorch, Transformers, RAG, LangChain, TensorFlow, Hugging Face, Vector DB, OpenAI | Pandas, NumPy, Scikit-learn
Backend & Frontend: Kafka, REST APIs, Microservices, React, Node.js, Nest.js, Zustand, Event-driven Architecture, TypeORM,
Data & Cloud: PostgreSQL, Redis, Git, Docker, CI/CD, Azure DevOps, AWS, Environment Configuration, Datadog

Education

Columbia University New York, NY
Master of Arts in Statistics September 2024 – December 2025
Core Curriculum: Generative AI using LLMs, Natural Language Processing, Advanced Machine Learning, Statistical Machine Learning, Algorithm for Data Science, Applied Data Science, Advanced Data Analysis, Linear Regression, Probability, Inference

The Chinese University of Hong Kong, Shenzhen Shenzhen, China
Bachelor of Science in Applied Mathematics September 2020 – July 2024
Core Curriculum: Machine Learning, Data Structures, Optimization, Stochastic Process, Mathematical Modeling

Work Experience

Antares Capital LP New York, NY
Software Engineer March 2026 – Present

- Re-architected legacy service from client-side full data loading to **server-side pagination**, replacing "load-all + in-memory filtering" with **database-driven SQL queries**. Reduced data transfer size from ~50MB per request to paginated responses (~200KB), improving page load time from 10–30s to ~1–2s.
- Designed and optimized database access patterns by introducing pagination-friendly **indexing strategies**, significantly improving query latency and stability under large datasets. Introduced **Redis-based distributed caching** to replace per-instance in-memory storages, enabling stateless services and horizontal scalability across multiple API servers.
- Collaborated with DBA team to support **database migrations** and ensure data consistency across environments. Diagnosed and resolved data staleness caused by misconfigured environment variables. Led post-migration stabilization by identifying and fixing downstream bugs introduced by architectural changes, improving system reliability.

Summoner New York, NY
AI Engineer April 2025 – December 2025

- Led the **research-to-product** progression by developing a focused **proof-of-concept** for stochastic multi-agent resume evaluation and generalizing those findings into a modular, production-ready s**calable machine learning evaluation and ranking system** for automated reasoning assessment with standardized logging, monitoring, and reproducibility controls.
- Architected an **LLM-driven multi-agent evaluation platform** that ingests user's context, generates domain-agnostic challenge questions with logical reasoning pathways, synthesizes AI personas, orchestrates real-time timed challenge sessions, and evaluates answers using **LLM-based logic extraction** and **embedding cosine similarity** scoring.
- Developed a **semantic evaluation engine** using GPT-4o for structured logic extraction and **OpenAI text-embedding-3-small** with cosine similarity scoring to automate partial-credit assessment. Designed a **configurable multi-factor scoring system** with dynamic weighting across reasoning complexity, difficulty tiers, and scenario variations.
- Engineered a **cost-observability AI safeguards** using tiktoken for preflight token counting; implemented real-time USD tracking and configurable budget ceilings to prevent runaway spending across concurrent LLM operations. Optimized **high throughput distributed LLM** operations using **asynchronous** Python (Asyncio) with semaphore-bounded concurrency.
- Designed an **inter-agent collaboration protocol** based on **state-machine flow engines**, enabling persona agents to autonomously request peer hints via asynchronous message queues, integrating guidance within bounded latency windows. Engineered an **extensible integration layer** using shared JSON schemas and configs for plug-and-play agent connections.
- Built a **SQL-based evaluation analytics pipeline** to collect agent telemetry and experimental data at scale, enabling **statistical diagnostics**, **performance benchmarking**, and iterative model optimization. Developed a Partial Least Squares regression **analytics dashboard** using cross-validated Q^2 metrics and VIP feature important scoring to identify key performance drivers across agent configurations, informing model weighting and architecture decisions.

Epsilla (YC 23) New York, NY
AI Product Strategist February 2025 – March 2025

- Built an **Ollama-powered** automated **RAG research system** using LangChain framework, integrating multi-source search (DuckDuckGo, Tavily, Perplexity), local LLM summarization, YouTube transcript ingestion, configurable loop, and SMTP email delivery. Delivered concise, citation-backed reports with improved research efficiency by 27%.
- Collaborated cross-functionally to **optimized AI agent use cases** across inbound sales, smart manufacturing. Designed a cohesive AI content strategy through competitive research on leading AI companies, raising click-through rates by 15%.